CSCE 420 – Homework #5
due: Sat, Dec 6, 2025, 11:59 pm

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1. Models in First-Order Logic.

Consider the following information about houses lining a street. Bob lives to the left of Alice. Carol lives to the left of Bob. David lives to the left of Bob. (Assume each person lives in a distinct house, and they all live on the same side of the street.) Furthermore, assume that the 'leftOf' relationship has the properties of a total order (antireflexive, symmetric, transitive). Write out axioms for 'leftOf' in FOL. Let the set of facts and axioms be S.

Axioms for leftOf:

- **Irreflexive:** $\forall x\ \neg leftOf(x, x)$
- **Asymmetric:** $\forall x \forall y\ (leftOf(x, y) \rightarrow \neg leftOf(y, x))$
- **Transitive:** $\forall x \forall y \forall z\ ((leftOf(x, y) \wedge leftOf(y, z)) \rightarrow leftOf(x, z))$
- **Total:** $\forall x \forall y\ (x \neq y \rightarrow (leftOf(x, y) \vee leftOf(y, x)))$
- **Facts:** leftOf(B, A), leftOf(C, B), leftOf(D, B)

(To keep things simple, you can use terms like 'A' to refer to Alice's house, 'B' for Bob's house, etc. That way, you don't have to have separate objects for people and houses.)

1a. Draw a model depicting the scene described. Call it M1, such that sat(M1,S) or M1$\in$models(S).

Because leftOf is a total order and we know C and D are to the left of B, and B is to the left of A, a possible ordering is:
C - D - B - A (left - right)
So, C is left of D, D is left of B, B is left of A.
This model is M1, and it satisfies all sentences in S.

1b. Write M1 out formally in terms of <U,D,R>

U = {c, d, b, a}
D: D(A) = a, D(B) = b, D(C) = c, D(D) = d
Relation R for leftOf: Since the order is $c < d < b < a$:
R(leftOf) = {(c, d), (c, b), (c, a),
(d, b), (d, a),
(b, a)}

1c. Draw a *different* model of S, M2.

Another valid ordering can be derived by just swapping C and D, since they are both to the left of B, and all of the facts would still hold:
D - C - B - A (left - right)
This model is M2, and it satisfies all sentences in S.

1d. Write out M2 as <U,D.R>
U = {c, d, b, a}
D: D(A) = a, D(B) = b, D(C) = c, D(D) = d
Relation R for leftOf: Since the order is d < c < b < a:
R(leftOf) = {(d, c), (d, b), (d, a),
(c, b), (c, a),
(b, a)}

1e. Draw another model M3 in which there are at least 5 people living on the street.

To make a model with at least 5 people living on the street, we can add another person/house labeled E for Ethan. We can keep the original facts and total order, and place E anywhere as long as all the constraints hold.

Another possible ordering:
E - C - D - B - A (left - right)

Here, all the constraints still hold since B is to the left of A, and C and D are both to the left of B.

This is model M3, and it satisfies all sentences in S.

1f. Write M3 as <U,D,R>

U = {e, c, d, b, a}
D: D(A) = a, D(B) = b, D(C) = c, D(D) = d, D(E) = e
Relation R for leftOf: Since the order is e < c < d < b < a:
R(leftOf) = {(e, c), (e, d), (e, b), (e, a),
(c, d), (c, b), (c, a),
(d, b), (d, a),
(b, a)}

1g. Suppose we added another axiom defining 'rightOf' as the *opposite* of 'leftOf', to create the theory (set of sentences) S'. Write the new axiom for 'rightOf'.
- **Does S' |= rightOf(A,D)?**
- **Does S' |= rightOf(C,D)?**
- **Does S' |= ∃ x rightOf(x,A)?**

Use model theory arguments (reasoning about models) to explain your answers.

New axiom for rightOf: $\forall x \forall y$ (rightOf(x, y) $\leftrightarrow$ leftOf(y, x))

Does S' |= rightOf(A,D)?

- rightOf(A, D) is equivalent to leftOf(D, A) by our new axiom.
- From our facts, by leftOf(D, B), leftOf(B, A) and transitivity, we get:
  leftOf(D,B) $\wedge$ leftOf(B,A) $\Rightarrow$ leftOf(D,A)
- Because the order is total and acyclic, every model of S will make D be at the left of A. So leftOf(D, A) is true in all models of S.

Therefore, **S′ ⊨ rightOf(A,D)**. So **yes, it is entailed.**

Does S' |= rightOf(C,D)?

- rightOf(C,D) $\leftrightarrow$ leftOf(D,C)
- In model M1, we see that C < D, or C is to the left of D. So leftOf(C, D) is true and leftOf(D, C) is false, making **rightOf(C, D) false in M1.**
- In model M2, we see that D < C, or D is to the left of C. So leftOf(D, C) is true, so **rightOf(C, D) is true in M2.**

Since there is at least one model where rightOf(C, D) is false (M1), it is not true in all models. Therefore, **no, it is not entailed.**

Does S' |= $\exists$ x rightOf(x,A)?

- By our axiom rightOf(x, A) $\leftrightarrow$ leftOf (A, x), rightOf(x, A) means leftOf(A, x).
- So essentially this is asking, must there be someone to the right of Alice in every model?
- In all valid models of S', the facts only force A to be to the right of B, C, and D. However, they do not force that anyone is to the right of A.
- A total order allows A to be the rightmost house.
- For example, **in model M1** (C - D - B - A), **no x satisfies rightOf(x, A)**

Since there exists at least one model of S' where there is nothing to the right of A, **the statement is not entailed.**

2. Bayesian Inference

Consider two factors that influence whether a student passes a given test: a) being smart, and b) studying. Suppose 30% of students believe they are intrinsically smart. But since students do not know a priori whether they are smart enough to pass a test, suppose 40% of will study for it anyway. (assume Smart and Study are independent). The causal relationship of these variables on the probability of actually passing the test can be expressed in a conditional probability table (CPT) as follows:

Table 1 Joint probability table about factors affecting whether students pass a test

P(pass|Smart,Study)	¬smart	smart
¬study	0.2	0.7
study	0.6	0.95

prior probabilities: P(smart)=0.3, P(study)=0.4

a) **Write out the equation for calculating joint probabilities, P(Smart,Study,Pass).**

- Using independence of Smart and Study and the chain rule, we have:
  $= P(S, T, P) = P(S) \cdot P(T) \cdot P(P \mid S, T)$
  $= \mathbf{P(S, T, P) = P(S)\ P(T)\ P(P \mid S, T)}$

b) **Calculate all the entries in the full joint probability table (JPT) [a 4x2 matrix, like Fig 12.3 in the textbook; [Note: names of variables are capitalized, lower-case indicates truth value, e.g. 'pass' means Pass=T, and '-pass' means Pass=F.]**

Smart	Study	Pass (P)	¬Pass (¬P)
S	**T**	0.114	0.006
S	**¬T**	0.126	0.054
¬S	**T**	0.168	0.112
¬S	**¬T**	0.084	0.336

c) **From the JPT, compute the probability that a student is smart, given that they pass the test but did not study.**

- We want: $P(S \mid P, \neg T) = P(S, \neg T, P) / P(P, \neg T)$
- Denominator: $P(P, \neg T) = P(S,\neg T, P) + P(\neg S, \neg T, P) = 0.126 + 0.084 = 0.21$
- So, $P(S \mid P, \neg T) = 0.126 / 0.21 = 0.6$
- Answer: **P(Smart | pass ∧ ¬study) = 0.6**

d) From the JPT, compute the probability that a student did not study, given that they are smart but did not pass the test.

- We want: $P(\neg T \mid S, \neg P) = P(S, \neg T, \neg P) / P(S, \neg P)$
- Denominator: $P(S, \neg P) = P(S, T, \neg P) + P(S, \neg T, \neg P) = 0.006 + 0.054 = 0.06$
- So, $P(\neg T \mid S, \neg P) = 0.054 / 0.06 = 0.9$
- Answer: **P(¬study | smart ∧ ¬pass) = 0.9**

e) Compute the marginal probability that a student will pass the test given that they are smart.

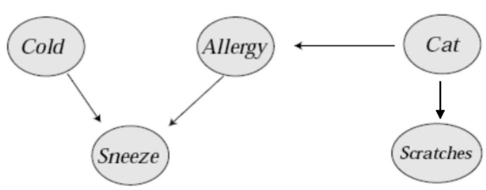
- We want: $P(P \mid S) = P(S, T, P) + P(S, \neg T, P) / P(S)$
- We know $P(S) = 0.3$
- Numerator: $P(S, T ,P) + P(S, \neg T, P) = 0.114 + 0.126 = 0.24$
- So, $P(P \mid S) = 0.24 / 0.3 = 0.8$
- Answer: **P(pass | smart) = 0.8**

f) Compute the marginal probability that a student will pass the test given that they study.

- We want: P(P | T)= P(S, T, P) + P(¬S, T, P) / P(T)
- We know P(T) = 0.4
- Numerator: P(S, T, P) + P(¬S, T, P) = 0.114 + 0.168 = 0.282
- So, P(P | T) = 0.282 / 0.4 = 0.705
- Answer: **P(pass | study) ≈ 0.705**

3. Bayesian Networks.

Here is a probabilistic model that describes what it might mean when a person sneezes, e.g. depending on whether they have a cold, or whether a cat is present and they are allergic. Scratches on the furniture would be evidence that a cat had been present.



Cold = False	Cold = True
.95	.05

Cat Presence = False	Cat Presence = True
.98	.02

Cat Presence	Allergic Reaction = False	Allergic Reaction = True
False	.95	.05
True	.25	.75

Cold	Allergic Reaction	Sneeze = False	Sneeze = True
False	False	.99	.01
False	True	.3	.7
True	False	.2	.8
True	True	.1	.9

Cat Presence	Scratches = False	Scratches = True
False	.95	.05
True	.5	.5

a) Using Equation 13.2 in the textbook (p. 415), write out the expression for the joint probability for any state (i.e. combination of truth values for the 5 variables in this problem). [Note: Use capital letters for names of variables, and lower-case to indicate truth value, e.g. 'cold' means Cold=T, and '-cold' means Cold=F.]

P(Cold,Sneeze,Allergic,Scratches,Cat) = **P(Cold) P(Cat) P(Allergic | Cat) P(Scratches | Cat) P(Sneeze | Cold, Allergies)**

b) Use the equation above to calculate the *joint probability* that the person sneezes, but does not have a cold, has a cat, is allergic, and there are scratches on the furniture:

P(-cold,sneeze,allergic,scratches,cat) = P(¬C) P(Cat) P(A | Cat) P(Sc | Cat) P(S | ¬C, A)

= 0.95 · 0.02 · 0.75 · 0.5 · 0.7
= 0.00498 ≈ **0.005**

c) Use *normalization* to calculate the conditional probability that a person has cat, given that they sneeze and are allergic to cats, but do not have a cold, and there are scratches on the furniture.

- P(cat|-cold,sneeze,allergic,scratches) = P(cat,E) / P(cat, E) + P(¬cat, E)
- P(cat,E)=P(¬Cold, Sneeze, Allergies, Scratches, Cat) ≈ 0.0049875
- P(¬cat, E) = P(¬C) P(¬Cat) P(A | ¬Cat) P(Sc | ¬Cat) P(S | ¬C, A) = 0.95 · 0.98 · 0.05 · 0.05 · 0.7 = 0.00162925
- P(cat, E) + P(¬cat, E) = 0.0049875 + 0.00162925 = 0.00661675
- So, **P(cat | ¬cold,sneeze,allergic,scratches)** = 0.0049875 / 0.00661675 = **0.754**

d) Use Bayes' Rule to re-write the expression for P(cat|scratches). Look up the values for the numerator in the table above.

- P(cat|scratches) = P(cat | Sc) = P(Sc | cat) P(cat) / P(Sc)
- P(Sc | cat)=0.5
- P(cat)=0.02
- Numerator = 0.5 · 0.02 = 0.01
- Denominator = P(Sc) = P(Sc | cat) P(cat) + P(Sc | ¬cat) P(¬cat) = 0.059
- So, **P(cat | scratches)** = 0.01 / 0.059 **= 0.17**

e) The denominator in the answer for (d) would require *marginalization* over how many joint probabilities? Write out the expressions for these (i.e. expand the denominator, but you don't have to calculate the actual values).

- The denominator in part D is P(scratches)
- If we compute it purely from the full joint distribution, we would marginalize over the other 4 binary variables which are **Cold, Allergic, Sneeze, Cat.**
- That means we would have 2^4 = **16 joint terms**

P(scratches) = P(cold,allergic,sneeze,cat,scratches)
+ P(cold,allergic,¬sneeze,cat,scratches)
+ P(cold,¬allergic,sneeze,cat,scratches)
+ P(cold,¬allergic,¬sneeze,cat,scratches)
+ P(¬cold,allergic,sneeze,cat,scratches)
+ P(¬cold,allergic,¬sneeze,cat,scratches)
+ P(¬cold,¬allergic,sneeze,cat,scratches)
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+ P(¬cold,allergic,sneeze,¬cat,scratches)
+ P(¬cold,allergic,¬sneeze,¬cat,scratches)
+ P(¬cold,¬allergic,sneeze,¬cat,scratches)

+ P(¬cold,¬allergic,¬sneeze,¬cat,scratches)

4. PDDL and Situation Calculus

To start a car, you have to be at the car and have the key, and the car has to have a charged battery and the tank has to have gas. Afterwards, the car will be running, and you will still be at the car and have the key after starting the engine.

a) **Write a <u>PDDL operator</u> to describe this action.**

(note: you can express this ego-centrically – you don't have to refer explicitly to the person starting the car; but the operator should take the car being started as an argument)

```
(:action start-car
 :parameters (?c - car)
 :precondition (and (at-car ?c)
                    (have-key ?c)
                    (battery-charged ?c)
                    (has-gas ?c))
 :effect (and (running ?c))
)
```

We stay at the car and keep the key because we can't delete those facts.

f) **b. Describe the same operator using <u>Situation Calculus</u> (remember to add a situation argument to your predicates).**

- Let the action be Start(c) and fluents AtCar(c, s), HaveKey(c, s), BatteryCharged(c, s), HasGas(c, s), Running(c, s)
- Precondition axiom: $\forall$c, s Poss(Start(c), s) $\leftrightarrow$ AtCar(c, s) $\wedge$ HaveKey(c, s) $\wedge$ BatteryCharged(c, s) $\wedge$ HasGas(c,s)
- Successor state axiom for Running: $\forall$c, a, s Running(c, do(a,s)) $\leftrightarrow$ (a = Start(c)) $\vee$ Running(c, s)
- Other fluents like AtCar and HaveKey are unchanged by Start, so their successor axioms just say that they persist.

g) **c. Add a <u>Frame Axiom</u> that says that starting this car will not changes whether any other car is out of gas (tank empty).**

- Let OutOfGas(c, s) mean car c's tank is empty in situation s.
- Starting car c_1 must not change the gas status of any other car $c_2 \neq c_1$
- $\forall$c1, c2, s (c1 $\neq$ c2) $\rightarrow$ (OutOfGas(c2, do(Start(c1), s)) $\leftrightarrow$ OutOfGas(c2 ,s))